

Well informed

Is green tea better for you than black tea?

Both offer health benefits, but the case for green tea is getting stronger

AFTER WATER, tea is the world's most popular beverage. Some drinkers may be attracted by its warmth on a cold day, others by the cobweb-clearing promise of its caffeine. But tea possesses health benefits, too. For one, it contains L-theanine, an amino acid thought to boost dopamine and serotonin, brain chemicals that lift mood. Tea is also rich in antioxidants that researchers have linked to reduced risks of developing cardiovascular disease and dementia.

Black teas (such as Assam and Darjeeling) account for more than two-thirds of consumption. Worldwide sales of green teas (such as matcha), however, have been steadily climbing in recent years, buoyed in part by reports that these may be healthier. But what does the science say?

One striking paper was published in *Molecules* in 2019. Its authors analysed data on 3,349 individuals aged 50 and up. After adjusting for smoking, coffee consumption and other variables, they found that those who drank green tea scored higher than black-tea drinkers on a "successful ageing" health index that assigned scores for such things as body weight, health and levels of physical and social activity.

One reason for this might be their respective L-theanine content. Green tea contains nearly 28% more per cup than black tea does, a study found in 2016. In

addition to promoting soothing brain chemicals, L-theanine has been shown by recordings of brain activity to amplify alpha brain waves—electrical oscillations that seem to sharpen mental focus without causing jitters. When consumed regularly, L-theanine seems to reduce stress and improve sleep. The benefits appear to increase with the daily dose, at least up to about 200mg, roughly the amount in eight cups of green tea.

Two studies bear this out. In one, published in *Nutrients* in 2019, 30 Japanese adults of all ages took 200mg of L-theanine daily for four weeks. They also spent another four weeks on a placebo. In addition to increasing verbal fluency and cognitive function, the supplements "significantly" improved sleep and reduced symptoms of anxiety and depression compared with placebo, the researchers wrote. In another trial, reported in the *Journal of Medicinal Food* in 2021, 26 Japanese volunteers aged 50 to 69 were given a supplement with only half as much L-theanine. After just a single dose, they performed notably better on tests of memory and reaction speeds than participants who had popped a placebo.

Green teas have also been shown to contain roughly a fifth more antioxidants than black ones. Although both are made from the leaves of the same plant, *Camellia sinensis*, black tea leaves are dried, crushed and left to darken as their compounds slowly react with oxygen. This oxidation converts many of tea's antioxidants into new molecules. According to a paper in the *International Journal of Molecular Sciences* in 2020, one antioxidant left behind in green tea eases inflammation and helps kill off damaged cells, possibly reducing risks of cardiovascular disease and cancers of the respiratory and digestive systems.

Black tea is not without its own benefits. It is richer than green tea in beneficial compounds such as theaflavins and thearubigins. Theaflavins reduce the body's absorption of cholesterol, a fatty molecule that can contribute to arterial plaque, whereas both theaflavins and thearubigins protect cells from oxidative and inflammatory damage. By all means make your next order a matcha, but do not chuck those black-tea bags just yet.

